

Accuracy of Lorenz number estimates based on transport data

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Introduction

- Thermal conductivity: $\kappa = \kappa_L$ (Lattice) + κ_e (electronic)
- Methods:
 - High $\vec{B}$ transport
 - Fitting over a model
- Typically used Wiedemann-Franz Law:
 $\kappa_L = \kappa - \kappa_e = \kappa - L\sigma T$;
 L : Lorenz number ($L_0 = 2.443 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$)

Expression for L

- L is not constant; depends on carrier density, scattering etc.
- Kim et al. proposed[†]:

$$L = 1.5 + \exp\left(-\frac{|S|}{116}\right)$$

Obtain L (in $10^{-8} \text{ W}\Omega\text{K}^{-2}$) from S (in μVK^{-1}).

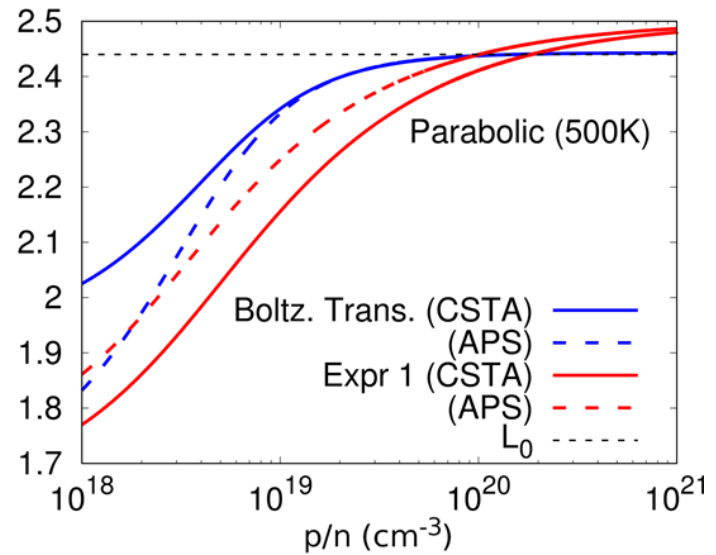
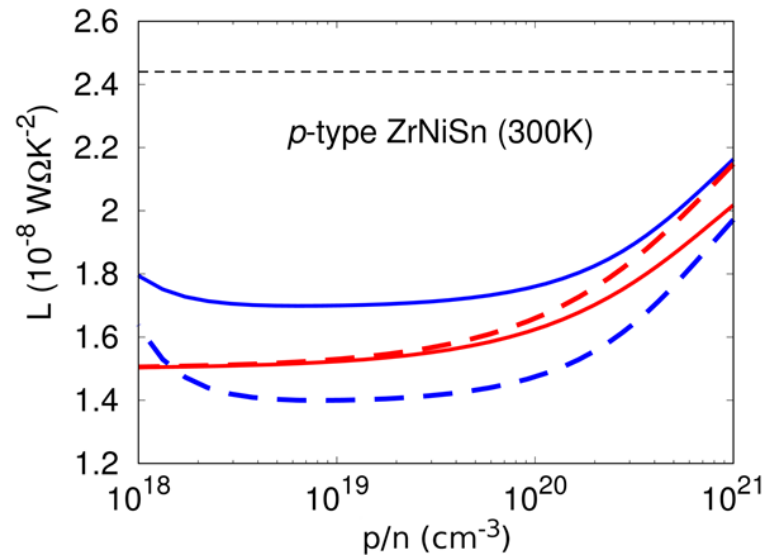
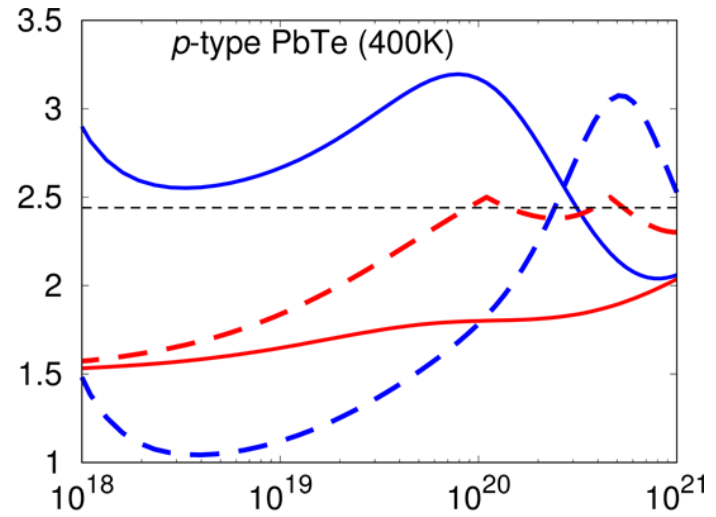
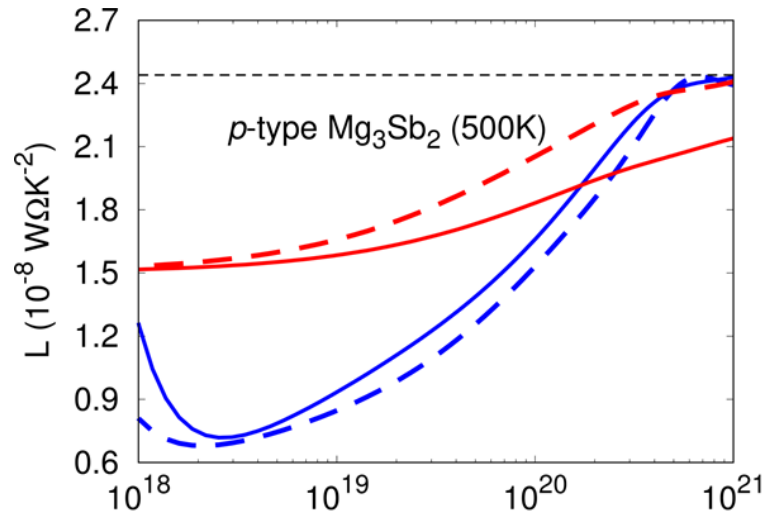
- Inspired from the single parabolic band, it utilizes the L vs S correlation.

[†] *APL Materials* **3**, 041506 (2015)

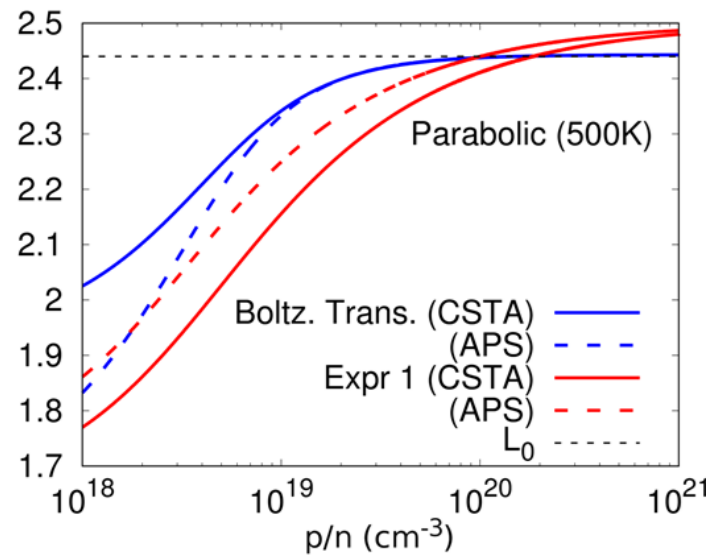
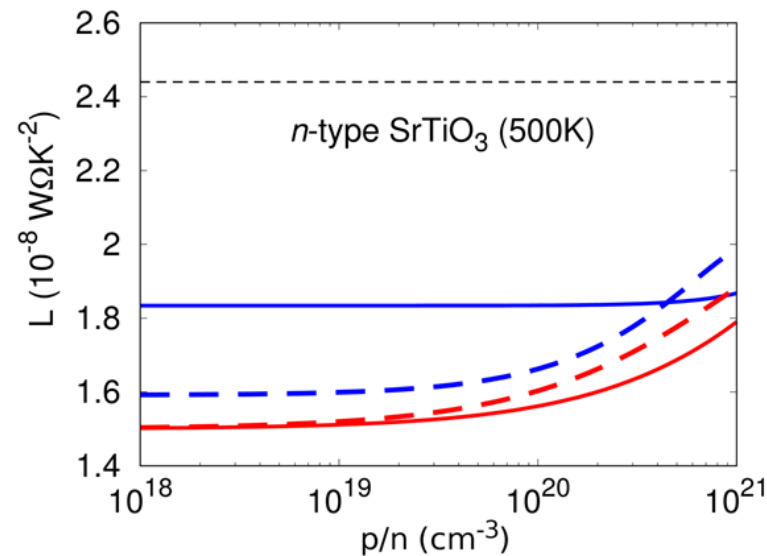
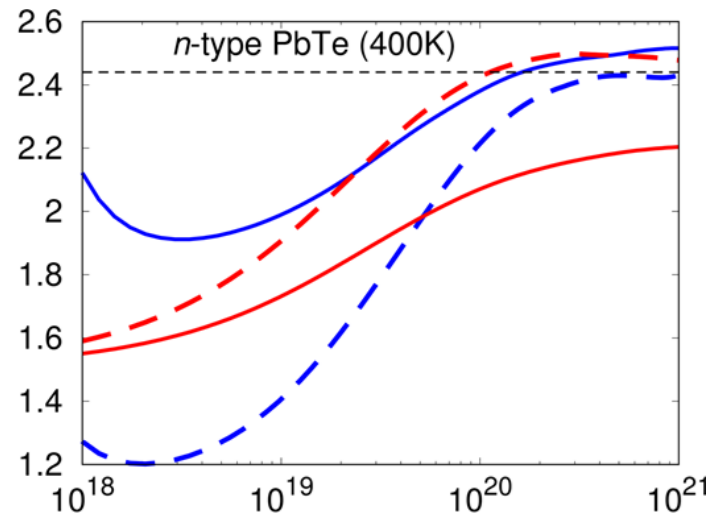
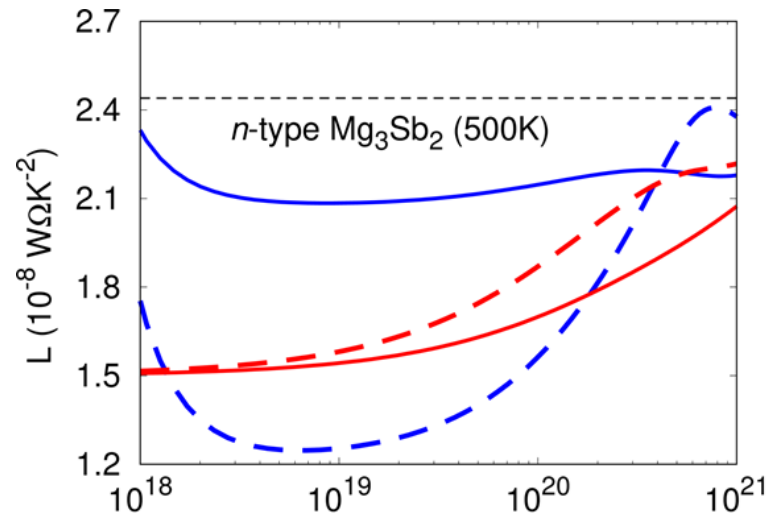
Methods

- Solved Boltzmann transport equations for L , S under two scattering time approx.:
 - CSTA; $\tau \propto E^0$
 - APS; or $\tau \propto 1/(T N(E))$ $N(E)$: DOS.
for parabolic bands: $\tau \propto (T E^{-1/2})$
- For thermoelectrics, large S causes a modification to the thermal conductivity such that:
$$L = L' - T\sigma S^2; \text{ where: } L' = (\kappa/\sigma T)$$

Results (p-type)



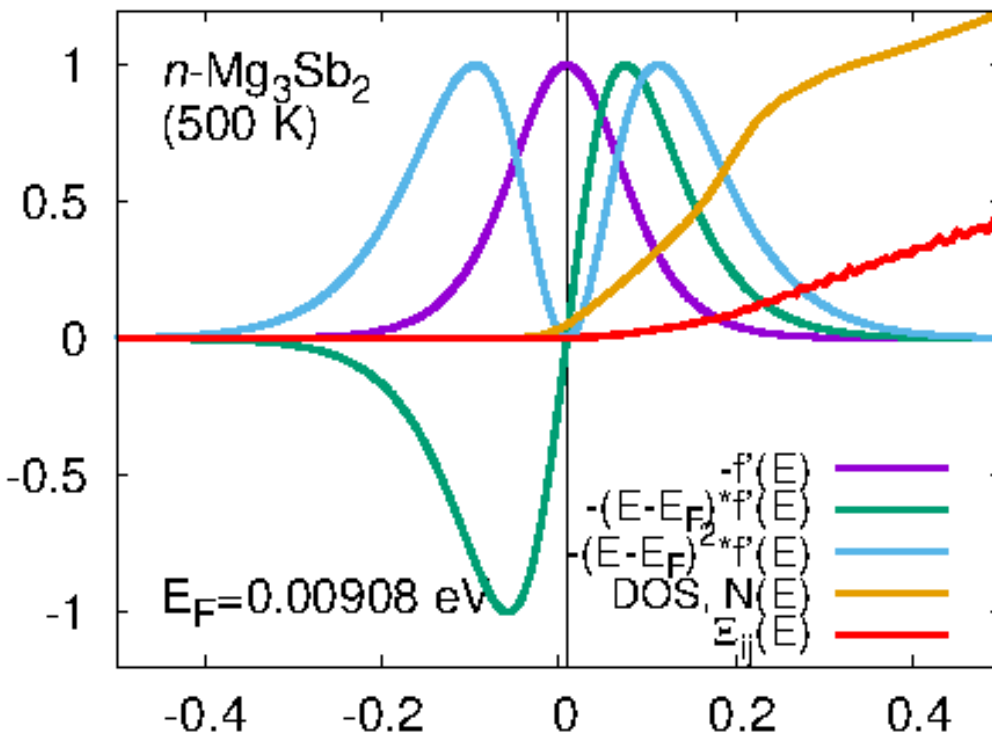
n-type



Observations

- Works alright with SrTiO_3 and ZrNiSn .
- Inaccurate with non-parabolic PbTe and especially Mg_3Sb_2 .
- Large deviations around ($n \sim 10^{19} \text{ cm}^{-3}$)
- Significant deviations with parabolic band model too.
- L doesn't converge to APS limit of 1.5×10^{-8} .

Window functions (at $5 \times 10^{19} \text{ cm}^{-3}$)



$$\sigma = \int \sigma(E) dE$$

$$S = -\frac{1}{eT} \frac{\int (E - E_F) \sigma(E) dE}{\int \sigma(E) dE}$$

$$\kappa_e = \frac{1}{e^2 T} \int (E - E_F)^2 \sigma(E) dE$$

where $\sigma(E) = e^2 \Xi(E) \left(-\frac{\partial f_0}{\partial E} \right)$

$$\Xi_{SPB}(E) = N(E) v(E)^2 \tau(E)$$

Conclusions

- The relationships between L and S are not direct, especially for non-parabolic bands, as shown.
- Care should be taken while approximating L using this approximation.

Results: Materials Today Physics **8** (2019) 49-55